

6. Particles X (of mass 4 units) and Y (of mass 9 units) move directly towards each other, collide and then separate.

If Δv_x is the change in velocity of X and Δv_y is the change in velocity of Y , the magnitude of ratio $\frac{\Delta v_x}{\Delta v_y}$ is

A 9 / 4

B 3 / 2

C 1

D 2 / 3

Answer: A

magnitude of change of momentum of X and Y equal, so $m_x \Delta v_x = m_y \Delta v_y \Rightarrow \frac{\Delta v_x}{\Delta v_y} = \frac{m_y}{m_x} = \frac{9}{4}$